

ME315 - Lab04

Matheus Rodrigues (259256)

1. Carregue os pacotes readxl e tidyverse.

```
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

2. Crie uma variável chamada fname que contenha o caminho para o arquivo WDIEXCEL.xlsx, disponível no seu diretório de dados. Confirme que o R identifica a existência do arquivo utilizando o comando file.exists.

```
fname <- "data/WDIEXCEL.xlsx"
file.exists(fname)
```

```
[1] TRUE
```

3. Leia a planilha “Country” e armazene no objeto countries.

```
countries <- read_excel(fname, sheet = "Country")
head(countries)
```

```
# A tibble: 6 x 31
  `Country Code` `Short Name`      `Table Name` `Long Name` `2-alpha code`
  <chr>          <chr>          <chr>        <chr>      <chr>
1 ABW           Aruba           Aruba        Aruba      AW
2 AFE           Africa Eastern and Sou~ Africa East~ Africa Eas~ ZH
3 AFG           Afghanistan      Afghanistan Islamic St~ AF
4 AFW           Africa Western and Cen~ Africa West~ Africa Wes~ ZI
5 AGO           Angola           Angola       People's R~ AO
6 ALB           Albania          Albania      Republic o~ AL
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <lgl>,
#   `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

4. Mantenha apenas os registros de 5 países (Brasil e outros 4 de sua escolha).

```
countries <- countries|>
  filter(`Short Name` %in% c("Brazil", "Angola", "Portugal", "Timor-Leste", "Mozambique"))

countries
```

```
# A tibble: 5 x 31
  `Country Code` `Short Name` `Table Name` `Long Name`      `2-alpha code`
  <chr>          <chr>      <chr>      <chr>      <chr>
1 AGO           Angola      Angola      People's Republic of ~ AO
2 BRA           Brazil      Brazil      Federative Republic o~ BR
3 MOZ           Mozambique Mozambique  Republic of Mozambique MZ
4 PRT           Portugal    Portugal    Portuguese Republic    PT
5 TLS           Timor-Leste Timor-Leste Democratic Republic o~ TL
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <lgl>,
#   `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

5. Leia a planilha “Data” e armazene no objeto WDI.

```
WDI <- read_excel(fname, sheet = "Data")
head(WDI)
```

```
# A tibble: 6 x 70
  `Country Name` `Country Code` `Indicator Name` `Indicator Code` `1960` `1961`
  <chr>          <chr>          <chr>          <chr>          <dbl> <dbl>
1 Africa Eastern~ AFE          Access to clean~ EG.CFT.ACCS.ZS      NA      NA
2 Africa Eastern~ AFE          Access to clean~ EG.CFT.ACCS.RU.~    NA      NA
3 Africa Eastern~ AFE          Access to clean~ EG.CFT.ACCS.UR.~    NA      NA
4 Africa Eastern~ AFE          Access to elect~ EG.ELC.ACCS.ZS      NA      NA
5 Africa Eastern~ AFE          Access to elect~ EG.ELC.ACCS.RU.~    NA      NA
6 Africa Eastern~ AFE          Access to elect~ EG.ELC.ACCS.UR.~    NA      NA
# i 64 more variables: `1962` <dbl>, `1963` <dbl>, `1964` <dbl>, `1965` <dbl>,
#   `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>, `1970` <dbl>,
#   `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>, `1975` <dbl>,
#   `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>, `1980` <dbl>,
#   `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>, `1985` <dbl>,
#   `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>, `1990` <dbl>,
#   `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, `1995` <dbl>, ...
```

6. Selecione apenas as colunas:

- Country Code
- Indicator Code
- E todos os anos de 1960-2018

```
WDI_filtered <- WDI |>
  select(`Country Code`, `Indicator Code`, `1960`:`2018`)
head(WDI_filtered)
```

```
# A tibble: 6 x 61
  `Country Code` `Indicator Code` `1960` `1961` `1962` `1963` `1964` `1965`
  <chr>          <chr>          <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 AFE          EG.CFT.ACCS.ZS      NA      NA      NA      NA      NA      NA
2 AFE          EG.CFT.ACCS.RU.ZS    NA      NA      NA      NA      NA      NA
3 AFE          EG.CFT.ACCS.UR.ZS    NA      NA      NA      NA      NA      NA
4 AFE          EG.ELC.ACCS.ZS      NA      NA      NA      NA      NA      NA
5 AFE          EG.ELC.ACCS.RU.ZS    NA      NA      NA      NA      NA      NA
6 AFE          EG.ELC.ACCS.UR.ZS    NA      NA      NA      NA      NA      NA
# i 53 more variables: `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>,
#   `1970` <dbl>, `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>,
```

```
# `1975` <dbl>, `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>,
# `1980` <dbl>, `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>,
# `1985` <dbl>, `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>,
# `1990` <dbl>, `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>,
# `1995` <dbl>, `1996` <dbl>, `1997` <dbl>, `1998` <dbl>, `1999` <dbl>, ...
```

7. Confirme se os dados wm WDI estão em formato tidy. Se não estiverem, transforme-os para tal formato. Faça de forma que exista uma coluna referente ao ano chamada YEAR.

Não está em formato tidy, pois os dados referentes aos anos estão dispostos em colunas distintas.

```
WDI <- WDI %>%
  pivot_longer(
    cols = `1960`:`2025`,
    names_to = "YEAR",
    values_to = "VALUE"
  )

head(WDI)
```

```
# A tibble: 6 x 6
  `Country Name`    `Country Code` `Indicator Name` `Indicator Code` YEAR  VALUE
  <chr>             <chr>          <chr>           <chr>          <chr> <dbl>
1 Africa Eastern a~ AFE            Access to clean~ EG.CFT.ACCS.ZS  1960     NA
2 Africa Eastern a~ AFE            Access to clean~ EG.CFT.ACCS.ZS  1961     NA
3 Africa Eastern a~ AFE            Access to clean~ EG.CFT.ACCS.ZS  1962     NA
4 Africa Eastern a~ AFE            Access to clean~ EG.CFT.ACCS.ZS  1963     NA
5 Africa Eastern a~ AFE            Access to clean~ EG.CFT.ACCS.ZS  1964     NA
6 Africa Eastern a~ AFE            Access to clean~ EG.CFT.ACCS.ZS  1965     NA
```

8. Confirme se a coluna referente ao ano está em formato inteiro.

```
class(WDI$YEAR)
```

```
[1] "character"
```

```
WDI <- WDI|>
  mutate(YEAR = as.integer(YEAR))

head(WDI)
```

```
# A tibble: 6 x 6
  `Country Name`   `Country Code` `Indicator Name` `Indicator Code` YEAR VALUE
  <chr>            <chr>          <chr>           <chr>          <int> <dbl>
1 Africa Eastern a~ AFE            Access to clean~ EG.CFT.ACCS.ZS   1960    NA
2 Africa Eastern a~ AFE            Access to clean~ EG.CFT.ACCS.ZS   1961    NA
3 Africa Eastern a~ AFE            Access to clean~ EG.CFT.ACCS.ZS   1962    NA
4 Africa Eastern a~ AFE            Access to clean~ EG.CFT.ACCS.ZS   1963    NA
5 Africa Eastern a~ AFE            Access to clean~ EG.CFT.ACCS.ZS   1964    NA
6 Africa Eastern a~ AFE            Access to clean~ EG.CFT.ACCS.ZS   1965    NA
```

```
class(WDI$YEAR)
```

```
[1] "integer"
```

9. Para o objeto WDI, mantenha apenas os registros dos países listados em countries. Selecione as colunas:

- Country Code
- Short Name
- YEAR
- SP.DYN.CBRT.IN

```
WDI <- WDI |>
  semi_join(countries, by = "Country Code") |>
  filter(`Indicator Code` == "SP.DYN.CBRT.IN") |>
  left_join(
    countries %>% select(`Country Code`, `Short Name`),
    by = "Country Code"
  ) |>
  rename(`SP.DYN.CBRT.IN` = VALUE) |>
  select(`Country Code`, `Short Name`, YEAR, `SP.DYN.CBRT.IN`)

head(WDI)
```

```
# A tibble: 6 x 4
  `Country Code` `Short Name` YEAR SP.DYN.CBRT.IN
  <chr>          <chr>    <int>    <dbl>
1 AGO           Angola    1960     51.6
2 AGO           Angola    1961     52.0
3 AGO           Angola    1962     52.2
4 AGO           Angola    1963     52.3
5 AGO           Angola    1964     52.4
6 AGO           Angola    1965     52.4
```

10. Ordene o objeto de forma que, para cada país, as informações estejam ordenadas por ano.

```
WDI <- WDI |>
  arrange(`Country Code`, YEAR)

head(WDI)
```

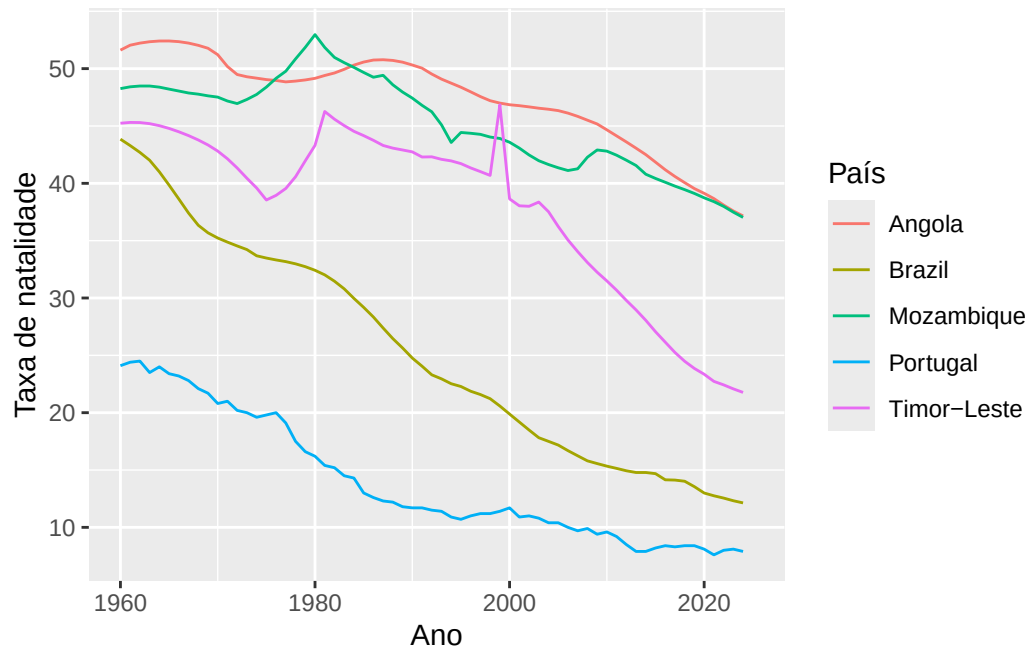
```
# A tibble: 6 x 4
  `Country Code` `Short Name` YEAR SP.DYN.CBRT.IN
  <chr>          <chr>      <int>      <dbl>
1 AGO           Angola      1960        51.6
2 AGO           Angola      1961        52.0
3 AGO           Angola      1962        52.2
4 AGO           Angola      1963        52.3
5 AGO           Angola      1964        52.4
6 AGO           Angola      1965        52.4
```

11. Construa um gráfico de linhas, usando o pacote ggplot2, que tenha no eixo X a variável YEAR, no eixo Y a variável SP.DYN.CBRT.IN e que cada linha corresponda a um país diferente.

```
library(ggplot2)

ggplot(WDI, aes(x = YEAR, y = `SP.DYN.CBRT.IN`, color = `Short Name`)) +
  geom_line() +
  labs(x = "Ano", y = "Taxa de natalidade", color = "País")
```

Warning: Removed 5 rows containing missing values or values outside the scale range (`geom\_line()`).



```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-09 20:02:44 -03"
```